

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)		
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Section / Batch:	IT - 4th year	Date:	01-08-2026

Code of Conduct

I K. Meghana 192321126 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Meghana

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10
GRAND TOTAL:			50 Marks

Annexure A – Class Test

1. Explain the concept of image enhancement in the spatial domain. Discuss how basic gray-level transformations, histogram processing techniques, and spatial filtering can be used to improve the quality of the captured images. Justify the suitability of each technique for this scenario.
2. Differentiate between image smoothing and image sharpening. Explain the working principles of spatial smoothing filters and sharpening filters. Discuss how these techniques improve image quality and explain the situations in which each technique should be applied.
3. Explain the image enhancement process in the frequency domain. Describe the roles of low-pass filters and high-pass filters in image enhancement. Compare their effects on image quality and discuss suitable applications for each filtering technique.
4. Explain the concept of image segmentation. Compare point detection, line detection, edge detection, thresholding, and region-based segmentation with respect to their working principles, advantages, limitations, and practical applications. Identify the most appropriate technique(s) for this scenario and justify your answer.
5. Explain the importance of edge detection, edge linking, and boundary detection in image processing. Discuss the working principles of these techniques and describe how they contribute to accurate object recognition and defect detection. Illustrate your answer with suitable examples and diagrams wherever necessary.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1/ Image enhancement in The Spatial Domain.

Def:

Image enhancement in the spatial domain improves the quality of an image by directly the pixel values.

Increases brightness, contrast & remove noise

* Gray-Level Transformation

→ It operate directly on pixel intensities.

→ It used to adjust brightness & contrast.

ex

* Image Negative

* log transformation

* ~~ROI~~

AdvAdvantages

→ simple and fast

→ Improves visibility of dark or bright regions.

Histogram Processing

→ Improves image contrast by redistributing gray levels.

Technique

- Histogram equalization

- Histogram specification

- Histogram stretching

Advantages

- Enhances contrast

- Reveals hidden details

Spatial Filtering

- uses a filter mask or kernel on neighboring pixels
- smoothing filters (Mean, Gaussian)
- sharpening filters (Laplacian)

Advantages

- Remove noise
- enhances edges

Sustain Suitability

* Gray level transformation - brightness

* Histogram Processing - brightness

* Spatial Filtering - contrast

Differentiate between image smoothing & image sharpening

②

Image smoothing	Image sharpening
→ Remove noise → Blurs image → uses low-pass filters → Reduces fine details → used before segmentation	→ Enhance edges → Highlight detail → uses high pass → Increases fine details → used for feature extraction

Spatial smoothing filters

- Mean
- Median
- Gaussian

→ replace each pixel values using neighbor pixel values

Sharpening filter

- Laplacian filter
- Sobel filter
- Prewitt filter

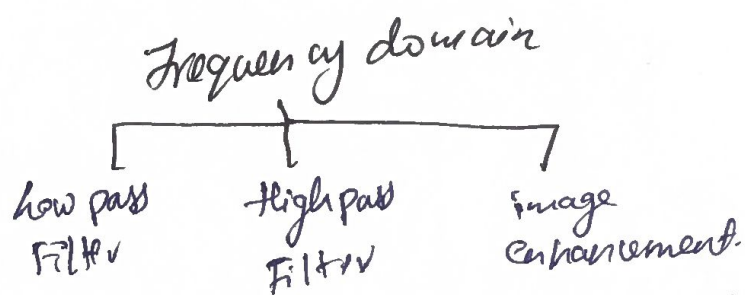
highlight sudden intensity changes to detect edges

Applications

Smoothing - Medical Images, noisy images

Sharpening - Satellite Images.

③ The Image Enhancement process in the Frequency domain
⇒ Frequency domain enhancement ~~transforms~~ ^{transforms} an image into the frequency domain using the Fourier Transform



low pass Filter

- Passes low frequencies
- Remove noise
- Produces smooth image

Application

- Medical Image
- Noise reduce

High pass Filter

- Passes high frequency
- enhances edges
- Produces sharper images

Application

- Object detection
- Fingerprint enhancement

Compare their effects on image quality.

- ~~Point detection~~ - ~~image defects~~
- ~~edge detection~~ - ~~object recognition~~
- ~~thresholding~~ - ~~document scanning~~.

low pass

- Removes noise
 - Smooth image
 - ~~It~~ Removes high frequency
- ### High pass
- Enhance edges
 - Sharp image
 - ~~It~~ Removes low frequency

④

Image Segmentation

It divides an image into meaningful region for easier analysis

Point detection

- Detects isolated pixels
- used for identifying defects.

Line Detection

- It Detects horizontal, vertical lines
- It used in road and document analysis.

Edge detection

- It detects object boundaries
- using sobel, Prewitt, Canny Methods.

Thresholding

- separates foreground & background based on intensity
- simple and fast

Region based segmentation

- Group neighboring pixels with similar properties
- produces connected regions

Point Detection - spot defect

Line Detection - Road detection

Edge detection - object recognition

Thresholding - Document scanning

Region based segmentation - Medical image segmentation

Edge Detection

- It detects sudden intensity changes representing object boundaries

using → Sobel
→ Prewitt
→ Canny
→ Roberts

Edge Linking

- It connects broken edge segments to form a continuous boundaries

using → Hough transform
→ Graph-based linking.

Boundary detection

* Finds the complete outer boundaries of any object

→ Accurate object recognition

→ Shape analysis

→ Defect detection

→ Medical image analysis

Examples

→ edge detection identifies object edges

→ edge linking joins broken edges

→ boundary detection outlines the complete object to detect defects.